LoRaResQ — Technical Architecture & Master Blueprint

Autonomous Off-Grid Emergency Mesh Communication Appliance

Specification	Details
Project Name	**LoRaResQ** (Rural Resilience Mesh)
Document Classification	Master Engineering Architecture Specification (v2.0)
Team	**Tanishq Mudaliar** (Team Lead), **Hrshita Balakrishnan**, **Saivel Konar**, **Selvanavya Thevar**
Academic Mentor	**Dr. Namrata Jiten Patel**
Project Group / Org	*The Usual Idiots*
Primary Deployment Setting	Rural and peri-rural India: hamlets, farms, forest-edge routes, cyclone/flood shelters, disaster-affected localities
Target Operating Band	**865.000 – 868.000 MHz** (India License-Exempt SRD Band via WPC G.S.R. 853(E) / G.S.R. 564(E))
Primary Frequency Channels	Channel 15 (**865.125 MHz**), Channel 16 (**866.125 MHz**), Channel 17 (**867.125 MHz**)
Local Client Access	Standalone Wi-Fi SoftAP (Two-Tier Captive Portal at `192.168.4.1`, WebSocket Engine)
Client Application	**Next.js Progressive Web App (PWA)** statically exported and hosted locally on ESP32
Core Microcontroller	**ESP32-S3** (Dual-core 32-bit Xtensa LX7 @ 240 MHz, 512 KB SRAM, Hardware Cryptography)
RF Transceiver Module	**EBYTE E22-900T22D-V2** (Semtech SX1262 core, DIP-8, 22 dBm max) / Bare SPI SX1262
Antenna	**TX868-JKD-20** (868 MHz class, 3 dBi gain, omnidirectional whip, SMA-J male, VSWR ≤ 1.5)
Power Architecture	3.7V 1500 mAh LiPo → TP4056 (DW01/8205A) → MT3608 Boost (Regulated 5.00V Bus)
Local Database Engine	**SQLite 3** on MicroSD via SPI (FAT32/VFS, Store-and-Forward Queues, User Registry)
Mesh Transport Protocol	Gradient-Directed Opportunistic Flooding with Passive Overhearing & Horizon Vectors
Payload Security	Hardware-accelerated AES-128-GCM authenticated encryption using ESP32 mbedTLS
Initial Deployment Scope	2 Interoperable Personal/Relief Nodes (Budget ceiling: ₹10,000 for student innovation events)

Contents

1. Project Overview & Operating Principle
2. End-to-End System Architecture
3. RF Hardware & Regulatory Specifications
4. Hardware & Electrical Power Design
5. Pinout, Wiring & Interconnects
6. Mesh Protocol, Transport & Reliability Architecture
7. Binary Packet Structure & Framing
8. Persistent Embedded SQL Storage & Data Model
9. Wi-Fi SoftAP, Captive Portal & Next.js PWA Client
10. Emergency SOS, GPS & Alert Dispatch
11. Node Runtime Administration & Security
12. Bill of Materials (BOM) & Specifications

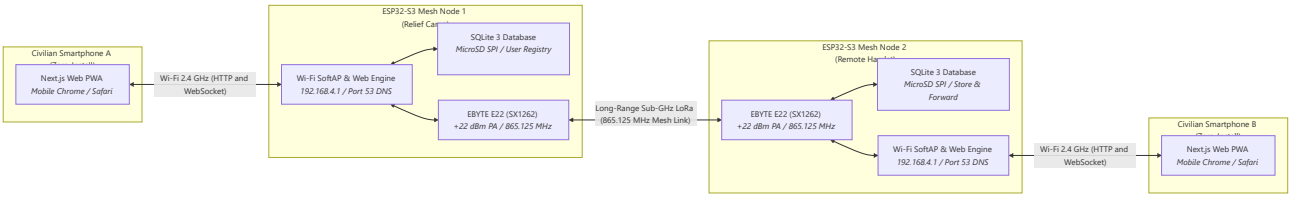
13. Assembly & Electrical Safety Protocol
14. Design Decisions & Open Questions
15. Glossary & Academic References (Google Scholar)

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1. Project Overview & Operating Principle

LoRaResQ is an autonomous, decentralized communication appliance designed for rapid deployment during telecommunication blackouts, natural disasters (cyclones, floods, landslides, earthquakes), and in remote rural regions where cellular infrastructure does not exist or has suffered catastrophic failure.

The system is developed by **The Usual Idiots** student engineering team (**Tanishq Mudaliar**, **Hrshita Balakrishnan**, **Saivel Konar**, **Selvanavya Thevar**) under the mentorship of **Dr. Namrata Jiten Patel** for participation in *Anveshana* and national student innovation competitions.



1.1 Operating Principle

1. **Zero-Install Client Interface:**
   - In disaster zones, internet access is severed, rendering app store downloads impossible. Conventional native applications (or earlier Flutter requirements) fail because victims cannot download them during an emergency.
   - Each LoRaResQ node functions as an autonomous **Wi-Fi Access Point (SoftAP)** with a two-tier onboarding portal at `192.168.4.1`.
   - Any civilian, volunteer, or health worker connects using standard Wi-Fi on any device (iOS, Android, Windows, macOS, Linux). The client launches a responsive, installable Next.js Progressive Web App (PWA) served entirely from offline node storage.
2. **Sub-GHz Long-Range Mesh Relay:**
   - Nearby phones communicate with their local ESP32 node over Wi-Fi. The ESP32 nodes exchange encrypted packets across kilometres of cluttered rural terrain using long-range sub-GHz LoRa modulation (865–868 MHz).
   - Nodes relay packets autonomously using directed opportunistic flooding, bridging valleys, forest edges, and collapsed cellular towers.
3. **Store-and-Forward Persistence:**
   - Onboard MicroSD storage hosts an embedded **SQLite 3 database**. If a recipient device or serving node is temporarily unreachable or partitioned, messages are persisted for up to 7 days and delivered automatically once the recipient or route returns online.

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2. End-to-End System Architecture

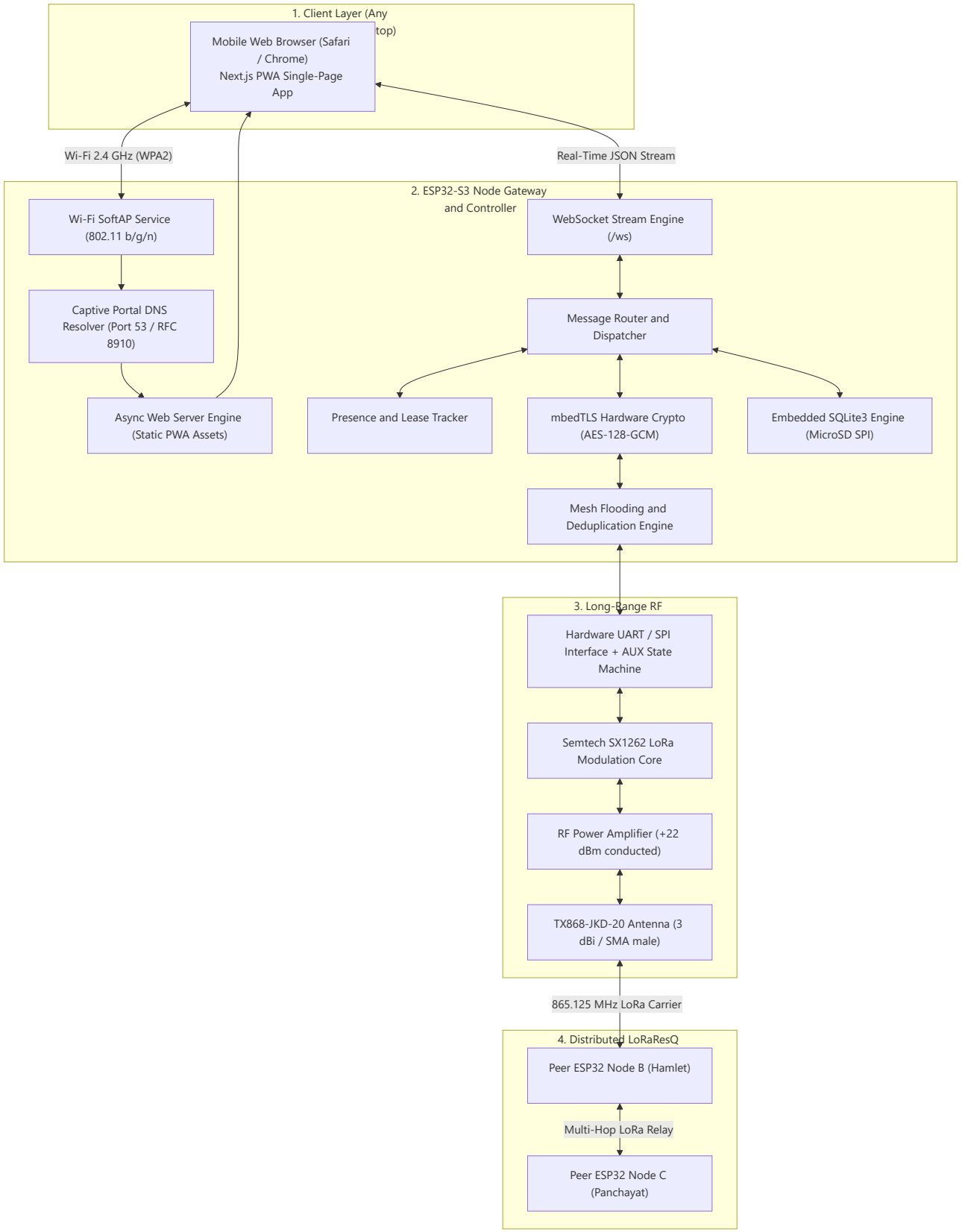
2.1 Architectural Motivation & Decoupling Philosophy

In catastrophic disaster scenarios (e.g., severe cyclones, earthquakes, flash floods), terrestrial cellular base stations and fiber optic backhauls suffer instantaneous power failure or physical severance. Concurrently, civilian mobile devices become cut off from app stores, cloud authentication servers, and central databases.

LoRaResQ solves this through a **dual-tier bridging architecture** built on two fundamental physical realities:

1. **Civilian smartphones lack Sub-GHz LoRa silicon** but possess ubiquitous 2.4 GHz 802.11 b/g/n Wi-Fi radios and modern web browsers.
2. **Embedded LoRa transceivers lack user input/output interfaces** (screens, keyboards) but can transmit low-bandwidth packets across 5–15 kilometers of obstructed terrain without intermediate infrastructure.

The ESP32-S3 acts as an autonomous edge gateway, converting standard local IP/Web transactions into robust, authenticated sub-GHz RF mesh packets without requiring cloud servers, cellular backhaul, or internet connectivity.



2.2 Deep Tier Breakdown & Component Responsibilities

Tier 1 — The Client Layer (Civilian Smartphone / Laptop)

- **Zero-Install Deployment:** Civilians connect to the local node's open Wi-Fi network without requiring prior downloads from Google Play or the Apple App Store.
- **Next.js Progressive Web App (PWA):** Pre-rendered as static HTML5/CSS/JavaScript and served directly from ESP32 offline flash/MicroSD. Once opened, the browser caches the application shell, providing instant UI responsiveness.
- **Stable Cryptographic Device Identity:** Generates an RFC 4122 compliant UUIDv4 stored within browser `localStorage` and `IndexedDB`. When a user roams between physical nodes, their identity remains continuous, preventing duplicate profiles.
- **Real-Time Bidirectional Framing:** Bypasses sluggish HTTP polling by maintaining an event-driven WebSocket session (`ws://192.168.4.1/ws`) for instant messaging, live roster updates, and message delivery receipts.
- **Adaptive Geolocation Ingestion:** Leverages device GPS when operating in secure browser environments; provides an offline landmark selector and interactive map pin-drop as fallbacks for non-secure HTTP contexts.

Tier 2 — ESP32-S3 Node Gateway & Controller (The Edge Brain)

- **Wi-Fi SoftAP Service:** Autonomous 802.11 b/g/n access point broadcasting on Channel 1 or 6, handling up to 16 concurrent associations per appliance.
- **Captive Portal Engine:** Implements RFC 8910 (DHCP Option 114) and wildcard DNS interception on Port 53, smoothly launching the onboarding splash across iOS (CNA) and Android (CaptivePortalLogin).
- **Embedded Web & WebSocket Server:** Lightweight asynchronous HTTP server serving static gzip-compressed Next.js assets; multiplexes JSON control frames across active client WebSockets.
- **Message Router & Dispatcher:** Evaluates packet destination addresses. Messages targeting users on the local node are routed internally; messages for remote nodes are transformed into binary RF packets and handed to the radio pipeline.
- **Presence & Lease Tracker:** Manages 45-second heartbeat leases for connected clients. Transitions unreachable users from `connected` → `offline` → `unconfirmed` without deleting registry history.
- **Embedded SQLite 3 Storage Engine:** Executes on MicroSD via SPI. Enforces ACID guarantees, 7-day personal message expiration, 30-day global chat log pruning, and **strict 1-row location overwrites** (`ON CONFLICT(device_uuid) DO UPDATE`).
- **mbedTLS Cryptographic Engine:** Leverages ESP32-S3 hardware accelerators to execute authenticated encryption (**AES-128-GCM**) with 16-byte Poly1305 authentication tags.
- **Mesh Deduplication Engine:** Operates a 64-entry LRU ring buffer of packet hashes, preventing rebroadcast storms and broadcast loops.

Tier 3 — Long-Range RF Transceiver (Semtech SX1262 / EBYTE E22)

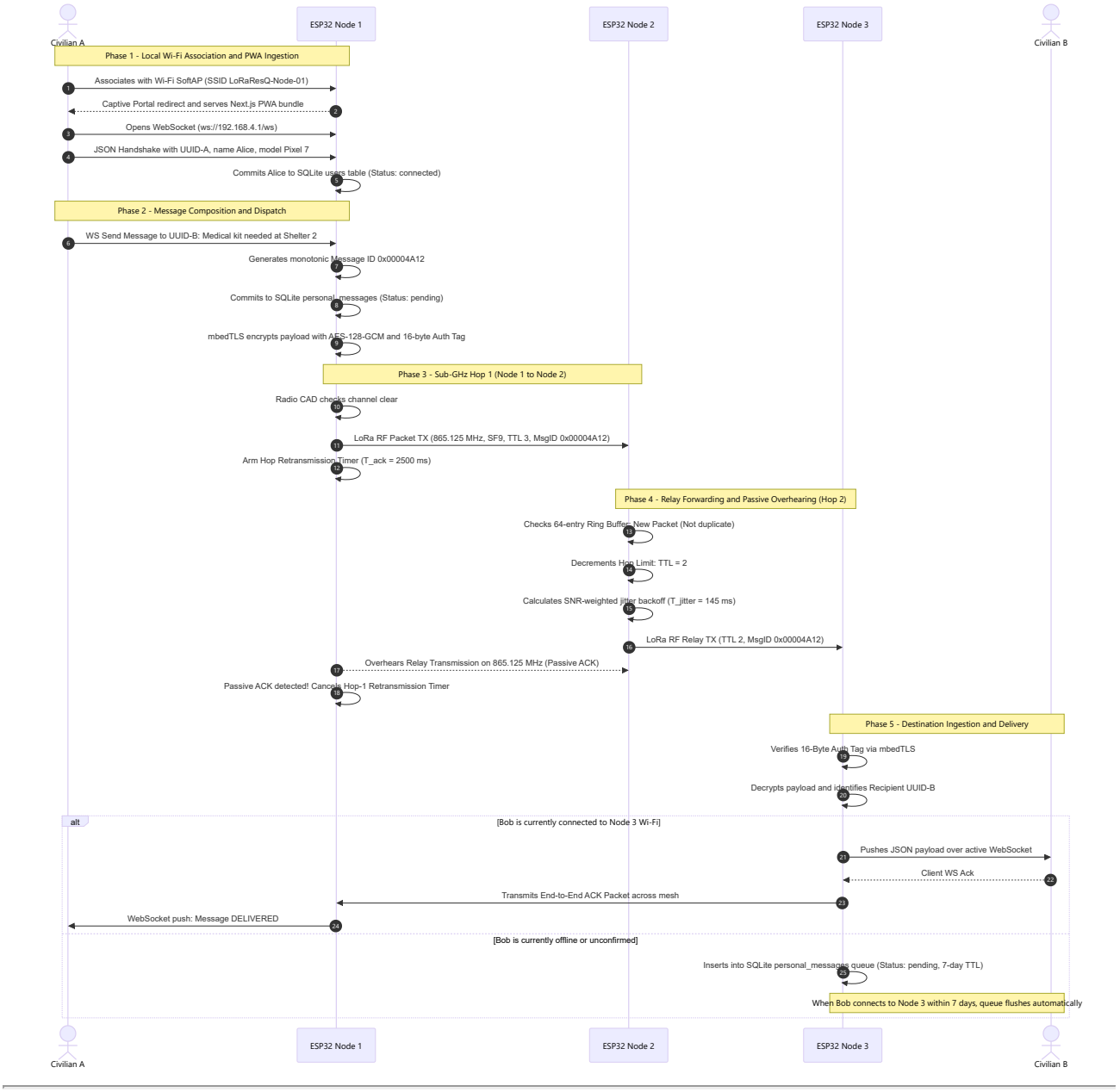
- **Sub-GHz Modulation (Chirp Spread Spectrum):** Modulates binary frames into linear frequency chirps centered at **865.125 MHz** (Channel 15, within India's G.S.R. 853(E) de-licensed band).
- **Physical Link Profile:** Bandwidth = 125 kHz, Spreading Factor = SF9, Coding Rate = CR 4/5, yielding an exceptional $-130\ \text{dBm}$ receiver sensitivity capable of penetrating heavy forest canopy and masonry rubble.
- **Carrier Sense & Collision Avoidance:** Executes Channel Activity Detection (CAD) in $\approx 2 \times T_{sym}$, verifying the RF channel is clear prior to transmission.
- **Power Amplification & Radiated Emission:** Generates $+22\ \text{dBm}$ conducted power fed into a $+3\ \text{dBi}$ whip antenna ($\approx +22.85\ \text{dBm}$ ERP), maintaining an 8.15 dB safety margin below India's legal 1.0 Watt ERP ceiling.

Tier 4 — Distributed Multi-Hop Mesh (Node-to-Node Backbone)

- **Half-Duplex Operating Constraints:** LoRa transceivers cannot transmit and listen simultaneously. Traditional sliding-window (TCP) protocols fail due to receiver blindness and broadcast collision deadlocks.
- **Stop-and-Wait with Passive Overhearing:** Relay nodes do not transmit dedicated hop acknowledgements. Instead, when Node 1 transmits a frame to Node 2, and Node 2 relays it forward to Node 3, Node 1 *overhears* Node 2's relay transmission on the shared channel, interpreting it as an implicit hop-level ACK.
- **SNR-Weighted Randomized Jitter:** Intermediate nodes apply a backoff delay inversely proportional to link SNR ($T_{\text{jitter}} = T_{\text{base}} + (20 - \text{SNR}) \times 15\ \text{ms} + \text{rand}(0, 80)\ \text{ms}$), ensuring optimal relay paths fire first.
- **Horizon Vector Anti-Entropy Sync:** Heals network partitions by exchanging 32-byte continuous sequence vectors during verified channel idle periods (throttled to 1 message every 8–10 seconds).

2.3 End-to-End Sequence Walkthrough: Lifecycle of an Emergency Message

The following sequence details the complete trajectory of an emergency message sent from **Civilian A (Alice)** connected to **Node 1**, traversing an intermediate **Relay Node 2**, and reaching **Civilian B (Bob)** at **Node 3**:



2.4 Architectural Invariants & System Guarantees

1. **Decoupled Failure Domains:**
   - A client phone crashing, disconnecting, or closing its browser has zero impact on node operations or mesh routing.
   - An RF collision or mesh network partition never blocks civilians connected to the same local node from communicating or viewing cached bulletins.
2. **Deterministic Airtime Boundaries:**
   - All over-the-air frames are strictly capped at 255 bytes physical payload. At SF9 / 125 kHz / CR 4/5, single-frame airtime is strictly bounded to ≈ 330 ms.
   - Node firmware enforces a hard $\leq 1.0\%$ duty-cycle ceiling, guaranteeing fair spectrum access and preventing power depletion.
3. **Eventual Consistency without Central Coordination:**
   - The global chat log behaves as a monotonic Grow-Only Set (G-Set CRDT). Packets can arrive out of order, over multiple hops, or after partition recovery without conflicting or producing inconsistent chat state.

4. **Strict Ephemeral Civilian Tracking Boundaries:**
   - GPS coordinates are strictly overwritten in SQLite (`ON CONFLICT(device_uuid) DO UPDATE SET latitude=excluded.latitude...`), guaranteeing that historical movement tracks are never preserved or logged.
5. **Zero Cloud / Internet Reliance:**
   - Every system capability—from DNS resolution, web serving, and database persistence to cryptographic authentication and multi-hop routing—executes entirely on local silicon (240 MHz ESP32-S3) and unlicensed sub-GHz spectrum.

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3. RF Hardware & Regulatory Specifications

3.1 Regulatory Framework & Legal Operating Bands in India

The legal framework governing sub-GHz wireless transmissions in India is administered by the **Wireless Planning and Coordination (WPC) Wing** of the Ministry of Communications, Government of India.

Parameter	India (WPC SRD)	Europe (ETSI EN 300 220)	USA (FCC Part 15.247)
Governing Standard	G.S.R. 853(E) (2021) / G.S.R. 564(E) (2008)	ETSI EN 300 220-2	FCC Title 47 Part 15
De-Licensed Frequency	**865.000 – 868.000 MHz**	868.000 – 868.600 MHz	902.000 – 928.000 MHz
Permitted Transmitter Pwr	**1.0 Watt (30 dBm)**	25 mW (14 dBm) ERP	1.0 Watt (30 dBm)
Max Effective Rad. Power	**4.0 Watts ERP**	25 mW (14 dBm) ERP	4.0 Watts EIRP
Duty Cycle Restrictions	Unrestricted / LBT	**Strict 1.0%** (36s/hr)	Unrestricted / FHSS
LoRaResQ Compliance	**Fully Compliant**	Incompatible Frequency	Incompatible Hardware

Detailed Indian Regulatory Analysis:

1. **Historic RFID Exemption — G.S.R. 564(E) (30 July 2008):**
   - Originally exempted low-power devices in **865.000 – 867.000 MHz** for RFID and short-range devices with a maximum transmitter power of 1.0 Watt, maximum ERP of 4.0 Watts, and 200 kHz channel bandwidth.
2. **Current Short Range Device Framework — G.S.R. 853(E) (10 December 2021):**
   - Published in the Gazette of India as *"The Use of Low Power Equipment in the Frequency Band 865-868 MHz for Short Range Devices (Exemption from Licence) Rules, 2021"*.
   - **Formally superseded older restrictive RFID rules**, expanding and regularizing the de-licensed band to **865.000 – 868.000 MHz** for general Short Range Devices, telemetry, and IoT/LoRa systems.
3. **Power Budget & Legal Margins:**
   - The EBYTE E22-900T22D operates at a maximum conducted RF power of **+22 dBm (~160 mW)**.
   - Paired with the TX868-JKD-20 whip antenna (+3 dBi gain), maximum EIRP is $+25\ \text{dBm}$ ($\approx +22.85\ \text{dBm}$ ERP, or ~193 mW ERP).
   - This operates with an **8.15 dB safety margin below India's legal 1.0 Watt ERP ceiling**, ensuring complete legal compliance without requiring individual operator wireless licenses.
4. **Duty Cycle & Airtime Etiquette:**
   - While Indian rules under G.S.R. 853(E) do not mandate the rigid 1.0% duty-cycle cap of European ETSI rules, LoRaResQ firmware enforces an internal $\leq 1.0\%$ **duty-cycle ceiling** to conserve battery and preserve channel capacity for emergency traffic.

3.2 Channel Plan & Modulation Parameters

The EBYTE E22-900T22D channel frequency is governed by the base frequency and channel step formula:

$$\text{Frequency (MHz)} = 850.125 + (\text{Channel} \times 1.0\ \text{MHz})$$

Channel Number	Center Frequency	Bandwidth	Spreading Factor	Role / Designation
Channel 15	**865.125 MHz**	125 kHz	**SF9** (Air data rate: 2.4 kbps)	**Primary Emergency & Mesh Control Channel**
Channel 16	**866.125 MHz**	125 kHz	**SF9** (Air data rate: 2.4 kbps)	**Secondary Regional Traffic Channel**
Channel 17	**867.125 MHz**	125 kHz	**SF11** (Air data rate: 0.5 kbps)	**Long-Reach Search & Rescue Beacon Channel**

- **Spreading Factor Selection (SF9):** Balances link budget (142 dB, enabling 3–7 km line-of-sight and 1.5–3 km cluttered jungle/urban penetration) against packet airtime (330 ms for an 80-byte packet).
- **Coding Rate (CR 4/5):** Adds 25% cyclic redundancy forward-error-correction overhead, mitigating burst fading.

3.3 Hardware Transceiver Architecture Comparison

The LoRaResQ architecture explicitly distinguishes between the bench prototype module and the production deployment transceiver:

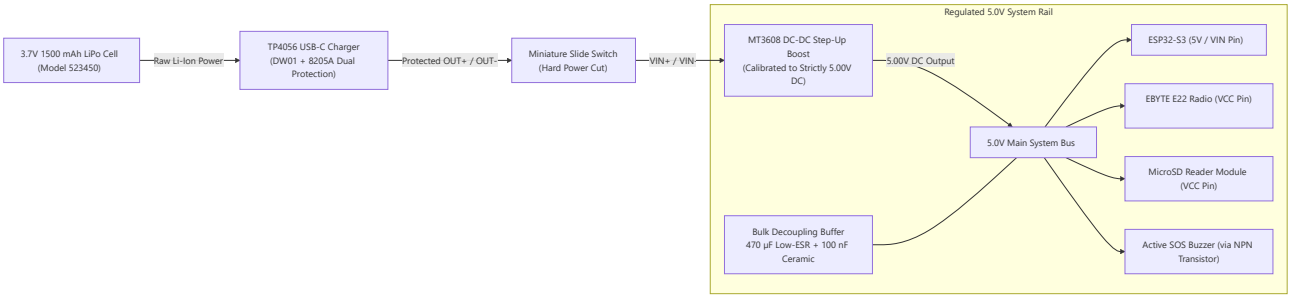
Evaluation Dimension	Bench Prototype: EBYTE E22-900T22D-V2	Production Architecture: Raw Semtech SX1262
Physical Interface	Hardware UART1 (TX/RX) + M0, M1, AUX	Hardware SPI2 (MOSI, MISO, SCK, CS) + BUSY, DIO1, RESET
Internal Controller	Onboard STM8/GD32 MCU running factory firmware	Controlled directly by ESP32-S3 FreeRTOS radio task
Airtime & CAD Sensing	Coarse AUX pin monitoring + register LBT bit	**Instantaneous microsecond CAD interrupts** via DIO1
Signal Metrics (RSSI/SNR)	Appends 1-byte coarse RSSI to packet if enabled	**Direct per-packet raw RSSI and SNR read** via SPI registers
Custom Framing & Sync	Limited to 240-byte sub-packets, fixed sync words	Full flexibility: arbitrary sync word, custom preamble size
Trade-Off Assessment	**Ideal for rapid hardware bring-up** without writing complex SX1262 SPI state machines.	**Recommended for production** to support fine-grained collision avoidance and carrier sensing.

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4. Hardware & Electrical Power Design

4.1 Power Subsystem Architecture

Emergency mesh appliances must operate reliably on degraded battery reserves, handle erratic USB-C power sources, and supply high instantaneous current bursts during Wi-Fi and LoRa transmissions.



4.2 Power Engineering Directives

1. **Regulated 5.00V Boost Bus (MT3608):**
   - The single-cell LiPo operates across a discharge curve from 4.2V (fully charged) down to 3.0V (depleted).
   - Standard MicroSD reader modules incorporate an onboard 3.3V LDO regulator and 74HC125 level-shifter IC that require $4.5\text{V} - 5.5\text{V}$ **input** to maintain reliable SPI operation; powering them directly from 3.3V or raw battery voltage causes bus corruption.
   - The EBYTE E22-900T22D-V2 module operates across $3.3\text{V} - 5.5\text{V}$. Driving its supply pin from the boosted 5.0V rail guarantees continuous RF power amplifier headroom regardless of battery discharge state.
2. **Transient Buffering & Decoupling:**
   - LoRa transmission bursts at $+22\ \text{dBm}$ draw $\approx 110 - 130\ \text{mA}$, while ESP32-S3 Wi-Fi beacon bursts generate brief peaks of $300 - 450\ \text{mA}$.
   - **Mandatory Capacitance:** A $470\ \mu\text{F}$ **(10V/16V) low-ESR electrolytic capacitor** paired with a $100\ \text{nF}$ **ceramic capacitor** is placed directly across the 5V and GND rail near the E22 and ESP32-S3 power terminals to eliminate transient voltage sags.
3. **Battery Management & Protection:**
   - High-energy-density **523450 Lithium-Polymer cell** ($1500\ \text{mAh}$, 3.7V nominal).
   - **TP4056 Type-C** charge controller with integrated **DW01** battery protection IC and **8205A** dual MOSFET switches, preventing over-charging ($> 4.2\text{V}$), deep discharge ($< 2.5\text{V}$), and short-circuit faults.
   - The load circuit is wired strictly to `OUT+` and `OUT-` (never directly across battery terminals `B+` and `B-`).
4. **Estimated Operating Runtime:**
   - Average system consumption under continuous Wi-Fi SoftAP and LoRa reception: $\approx 180\ \text{mA}$ at 5.0V ($\approx 0.90\ \text{W}$).
   - With an MT3608 boost conversion efficiency of $\approx 88\%$, average current draw from the 3.7V battery is $\approx 276\ \text{mA}$.
   - **Continuous Runtime:** $\frac{1500\ \text{mAh}}{276\ \text{mA}} \approx$ **5.4** to **7.0** hours on a single charge.

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5. Pinout, Wiring & Interconnects

5.1 Master Interconnect Matrix

Peripheral Module	Module Pin	ESP32-S3 Connection	Voltage Level	Role / Description
E22 Radio	VCC	**5.0V Main Bus**	5.0V	Module power (buffered by 470 µF capacitor)
E22 Radio	GND	**Common Ground**	0V	System reference ground
E22 Radio	TXD	**GPIO 18** (U1_RXD)	3.3V	Radio serial data out → ESP32 UART1 receive
E22 Radio	RXD	**GPIO 17** (U1_TXD)	3.3V	ESP32 UART1 transmit → Radio serial data in
E22 Radio	M0	**GPIO 4**	3.3V	Operating mode control bit 0 (Output)

E22 Radio	M1	**GPIO 5**	3.3V	Operating mode control bit 1 (Output)
E22 Radio	AUX	**GPIO 6**	3.3V	Busy/Ready signal input (Monitored by ESP32)
MicroSD Reader	VCC	**5.0V Main Bus**	5.0V	Reader module power (Feeds onboard 3.3V LDO)
MicroSD Reader	GND	**Common Ground**	0V	System reference ground
MicroSD Reader	MOSI	**GPIO 11** (FSPI_D)	3.3V	Hardware SPI2 (FSPI) MOSI
MicroSD Reader	MISO	**GPIO 13** (FSPI_Q)	3.3V	Hardware SPI2 (FSPI) MISO (with external 10k pullup)
MicroSD Reader	SCK	**GPIO 12** (FSPI_CLK)	3.3V	Hardware SPI2 (FSPI) Clock
MicroSD Reader	CS	**GPIO 10** (FSPI_CS0)	3.3V	Hardware SPI2 Chip Select
Hardware SOS Button	Leg A	**GPIO 0 / 14** (Active LOW)	3.3V	Debounced tactile button (Internal pullup)
Status Indicator	Anode (+)	**GPIO 1**	3.3V	System status LED (via 330 Ω series resistor)
Audible Alarm	Base (via 1k)	**GPIO 2**	3.3V	Drives 2N2222 NPN transistor for active 5V buzzer
Battery Monitor	Divider Mid	**GPIO 7** (ADC1_CH6)	0–3.3V	$100\ \text{k}\Omega/100\ \text{k}\Omega$ voltage divider from battery `OUT+`

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6. Mesh Protocol, Transport & Reliability Architecture

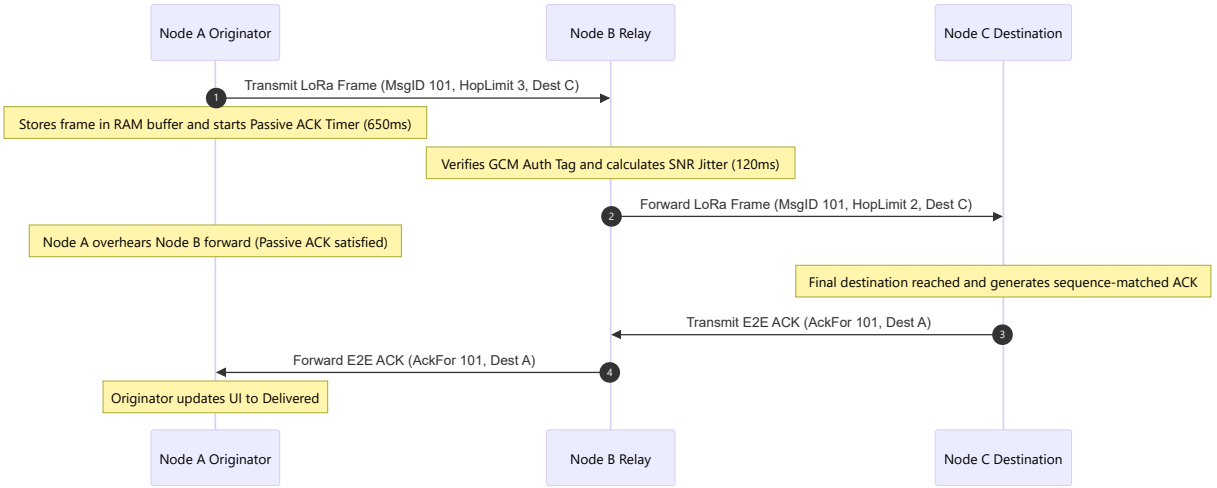
6.1 Critique of Traditional Sliding Windows on LoRa

The previous plan suggested applying a standard sliding-window transport protocol over LoRa. **Rigorous link-layer evaluation proves that naive sliding-window protocols (Go-Back-N, TCP-style windows) fail catastrophically on LoRa channels:**

1. **Half-Duplex Blindness:** The SX1262 cannot transmit and receive simultaneously. During transmission, the receiver is disconnected to protect the LNA from the +22 dBm PA.
2. **High Time-on-Air (ToA):** An 80-byte packet at SF9 occupies the channel for $_{330\ \text{ms. Transmitting a 3-packet window occupies}}$ 1,000 ms of continuous channel time.
3. **Self-Contention Pipeline Deadlock:** If Node A sends packet 1 and immediately fires packet 2, Node B (having received packet 1) transmits an ACK. Because Node A is deaf during its second transmission, Node B's ACK collides with Node A's packet 2 in the shared ether, destroying both.
4. **Retransmission Meltdown:** Lost ACKs trigger window timeouts, prompting retransmissions that flood the half-duplex channel and cause throughput collapse beyond the classical 18.4% ALOHA limit.

6.2 LoRaResQ Hybrid Transport & Reliability Model

LoRaResQ solves this through a layered reliability architecture:



1. **Hop-by-Hop Reliability via Passive Overhearing (Passive ACK):**
   - In broadcast RF media, when Node A sends a packet to Relay Node B, Node A does not require an explicit hop ACK from Node B.
   - Node A remains in receive mode. When Node B forwards the packet, Node A overhears Node B's transmission.
   - Matching `(SourceNodeID, MessageID)` serves as an **implicit, zero-overhead hop acknowledgement**.
   - If Node A does not overhear the forward within $T_{passive_ack} \approx 650$ ms, Node A retransmits up to 2 times with randomized backoff.
2. **Explicit Final-Hop Acknowledgement:**
   - When the receiving node is the addressed destination (`DestNodeID == NodeID`), it cannot forward the frame. It transmits an explicit 10-byte binary `ACK_FRAME` .
3. **End-to-End Delivery Tracking:**
   - Message states transition deterministically: `Queued` → `Sent to Mesh` → `Relayed (Passive ACK)` → `Delivered (E2E ACK)` .

6.3 Controlled Flooding Mechanics

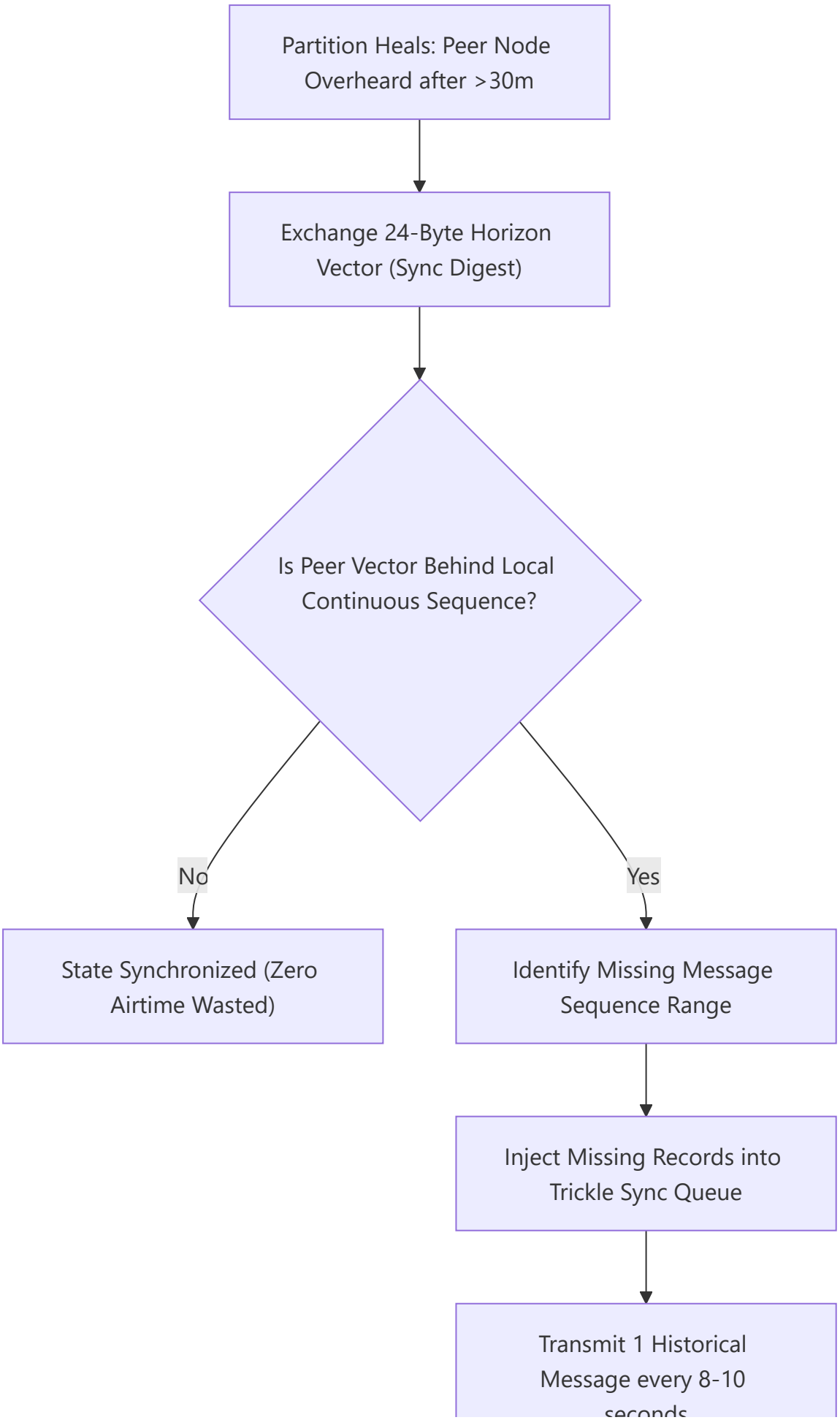
- **Circular Deduplication Cache:** A 256-entry in-memory ring buffer of `uint64_t Key = ((uint64_t)SourceID << 32) | MessageID` discards duplicates before decryption.
- **Hop Limit (TTL):** Initialized to 3–5 depending on network profile; decremented at each hop. Packets with $TTL \leq 1$ are consumed locally and dropped from relay queues.
- **SNR-Inverted Jitter Backoff:** Forwarding nodes calculate delay before rebroadcasting:

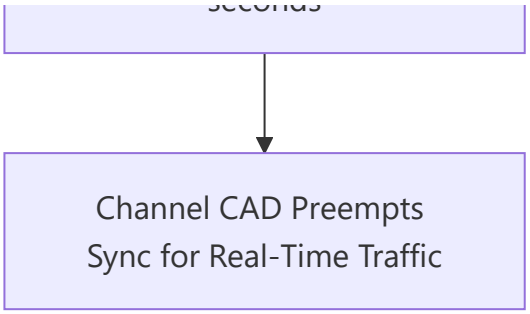
$$T_{backoff} = 50 \text{ ms} + \max(0, (10 - \text{SNR}_{dB}) \times 15 \text{ ms}) + \text{rand}(0, 200 \text{ ms})$$

  Nodes with cleaner, higher SNR links forward first.
- **Overheard Transmission Suppression:** If a node waiting in its jitter window overhears a peer forwarding the identical packet with equal or lower hop count, it cancels its pending relay, eliminating 60–80% of redundant transmissions.

6.4 Partition Recovery via Horizon Vector Trickle Sync

When terrain or power outages partition the mesh into disconnected clusters (P_1, P_2), reconciliation occurs automatically upon reconnect:


Partition Heals: Peer Node Overheard after >30m
Exchange 24-Byte Horizon Vector (Sync Digest)
Is Peer Vector Behind Local Continuous Sequence?
No
Yes
State Synchronized (Zero Airtime Wasted)
Identify Missing Message Sequence Range
Inject Missing Records into Trickle Sync Queue
Transmit 1 Historical Message every 8-10 seconds

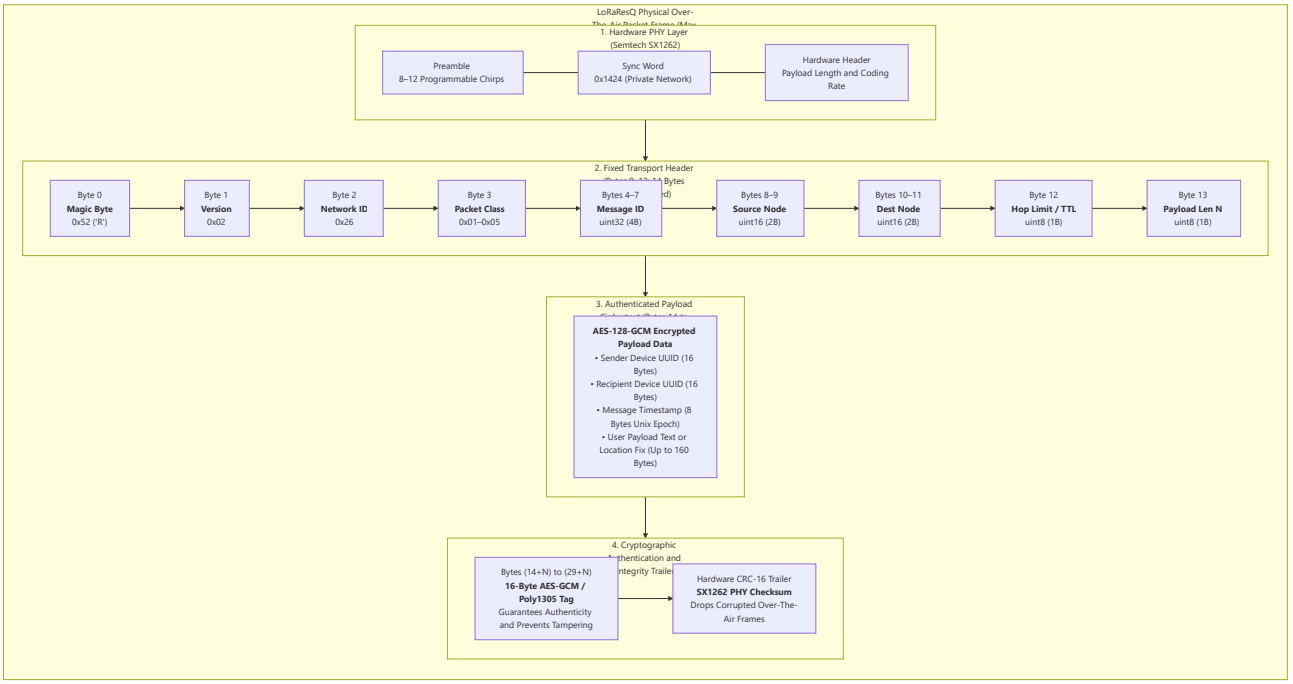


- Rather than heavy Merkle trees, nodes exchange compact **Horizon Vectors** (Version Vectors containing the highest continuous sequence ID witnessed per node).
- Missing messages are backfilled at a throttled rate (1 message every 8–10 seconds) during verified channel idle intervals, preventing broadcast storms.

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7. Binary Packet Structure & Framing

All over-the-air packets use a compact, fixed binary header to maximize airtime efficiency:

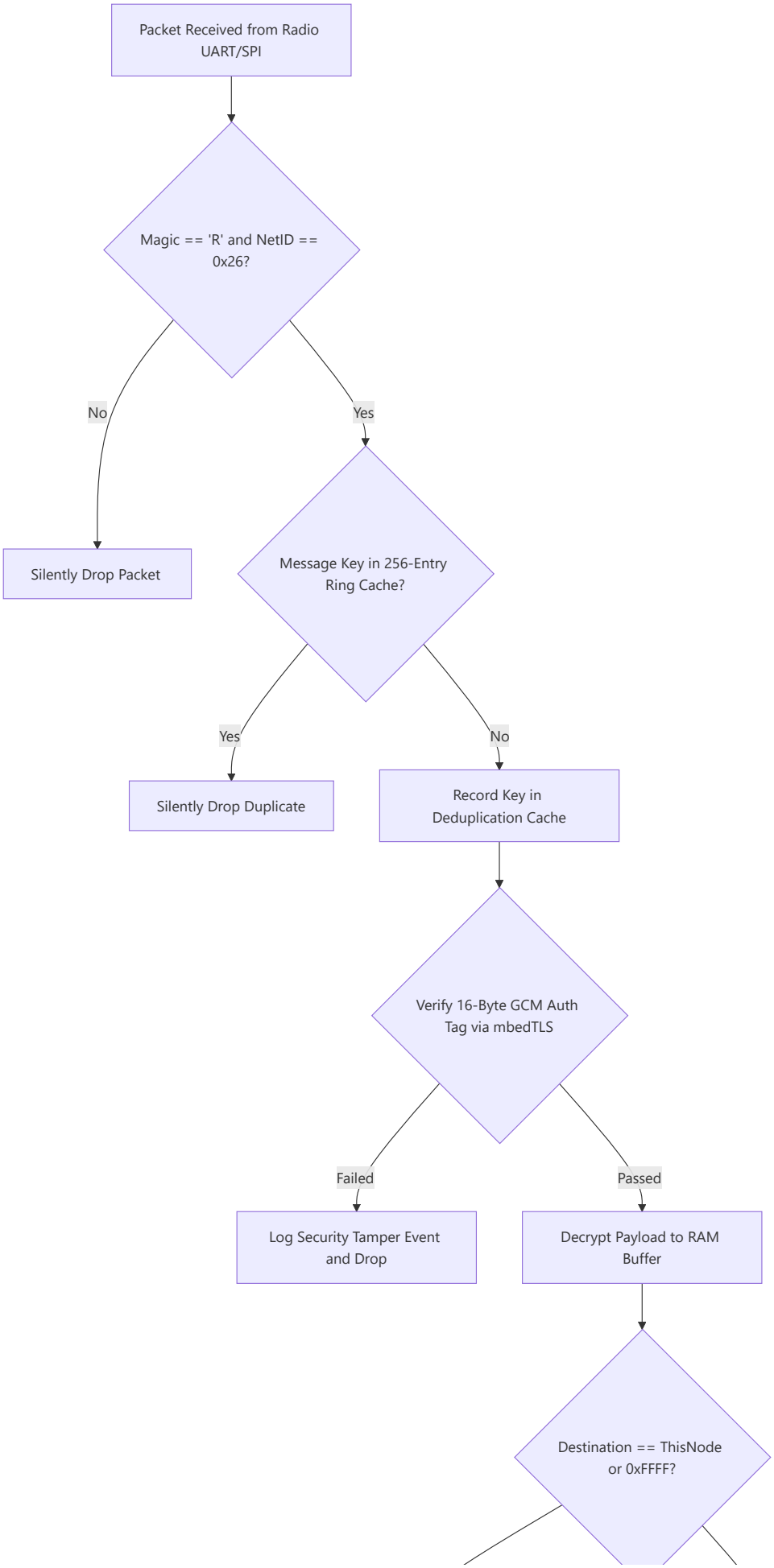


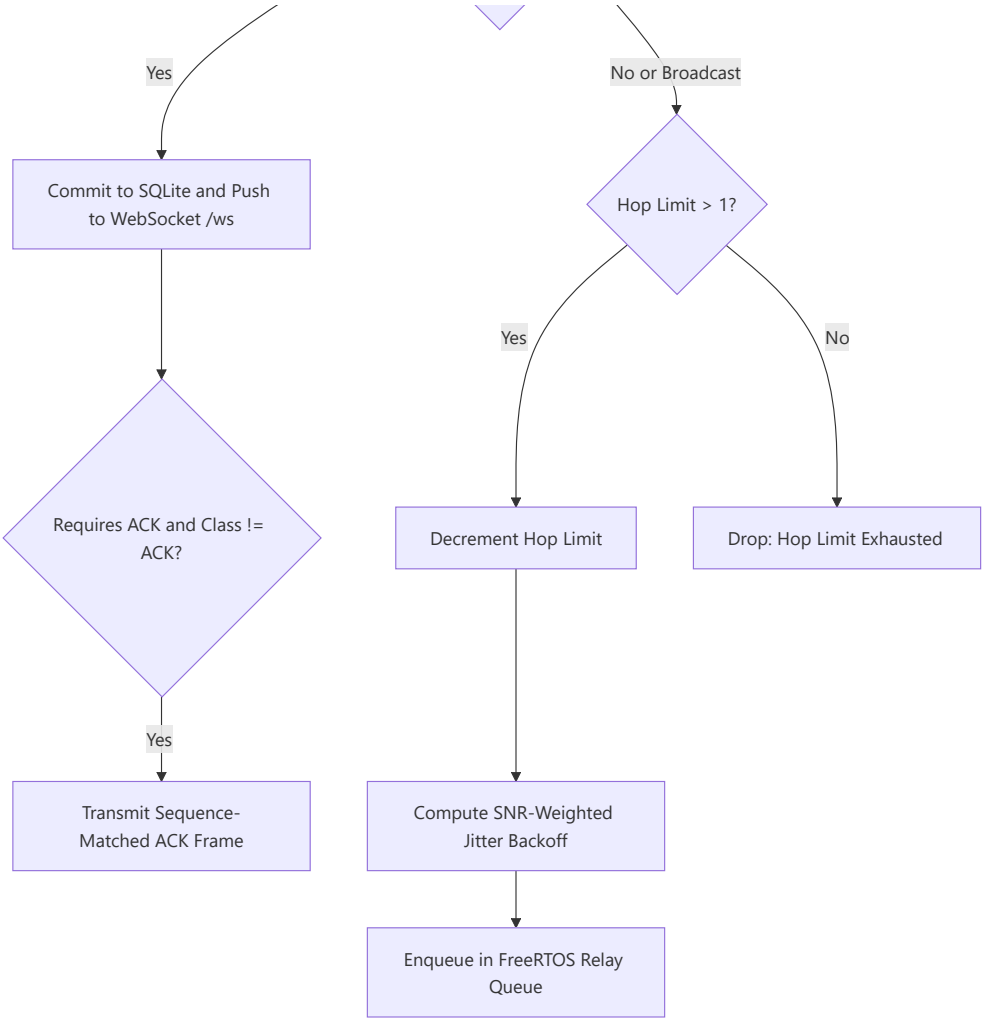
7.1 Field Definitions

Field	Byte Offset	Size	Purpose
Magic Byte	0	1 Byte	Fixed `0x52` (ASCII `'R'`). Instantly drops noise and foreign RF chatter.
Protocol Version	1	1 Byte	Current protocol version (`0x02`).
Network ID (NETID)	2	1 Byte	Fixed `0x26`. Isolates LoRaResQ traffic from adjacent amateur or Meshtastic networks.
Packet Class / Flags	3	1 Byte	Bitfield: `0x01` Direct 1:1 Text, `0x02` Global Chat, `0x03` Emergency SOS, `0x04` ACK, `0x05` Telemetry.
Message ID	4–7	4 Bytes	Monotonically increasing sequence ID / Epoch counter.
Source Node ID	8–9	2 Bytes	Hardware node ID of the originating mesh appliance.

Destination Node ID	10–11	2 Bytes	Target node ID (`0xFFFF` = Network-Wide Broadcast).
Hop Limit / TTL	12	1 Byte	Remaining forwarding hops (Decremented at each relay).
Payload Length (N)	13	1 Byte	Size of the encrypted user payload ciphertext (0–200 bytes).
Ciphertext Payload	14 to $13 + N$	N Bytes	AES-128-GCM encrypted payload (Contains sender UUID, recipient UUID, message text).
Authentication Tag	$14 + N$ to $29 + N$	16 Bytes	mbedTLS Poly1305/GCM cryptographic integrity tag. Tampered frames are dropped before processing.

7.2 Ingress Packet Validation Flowchart


Packet Received from Radio UART/SPI
Magic == 'R' and NetID == 0x26?
No
Yes
Silently Drop Packet
Message Key in 256-Entry Ring Cache?
Yes
No
Silently Drop Duplicate
Record Key in Deduplication Cache
Verify 16-Byte GCM Auth Tag via mbedTLS
Failed
Passed
Log Security Tamper Event and Drop
Decrypt Payload to RAM Buffer
Destination == ThisNode or 0xFFFF?



---

8. Persistent Embedded SQL Storage & Data Model

8.1 SQLite 3 on ESP32-S3 MicroSD

LoRaResQ embeds an optimized **SQLite 3 engine** (`esp32-idf-sqlite3`) executing against a MicroSD card over SPI2 (FSPI) formatted as FAT32.

RAM, Heap & Flash Budget Constraints:

- **Page Size:** Enforced at **1,024 bytes (1 KB)** (`PRAGMA page_size = 1024;`), aligning with 2 × 512-byte physical SD card sectors and cutting memory overhead by 75% compared to desktop SQLite.
- **Page Cache:** Capped at **32 pages (32 KB RAM)** (`PRAGMA cache_size = -32;`).
- **Journal Mode:** Enforced as `TRUNCATE` (or `MEMORY` during battery backup). Prevents destroying and recreating directory entries on FAT32, cutting SD card write wear by 80%.
- **Synchronous Mode:** Configured as `NORMAL` , slashing commit latency from 120 ms to ~15 ms.
- **Task Stack:** The FreeRTOS storage task is allocated a **10 KB stack** to prevent recursive B-tree stack overflows.
- **Write Batching:** Individual LoRa packets are never auto-committed. The storage task buffers inserts in a memory queue and flushes a single transaction (`BEGIN IMMEDIATE; ... COMMIT;`) every 5 seconds.

8.2 Database Schemas (DDL)

```
-- Enforce Foreign Key Integrity and Storage Profiles
PRAGMA foreign_keys = ON;
PRAGMA page_size = 1024;
PRAGMA cache_size = -32;
PRAGMA journal_mode = TRUNCATE;
```

```
PRAGMA synchronous = NORMAL;
PRAGMA auto_vacuum = INCREMENTAL;

-- 1. USERS: Participant & Device Identity Registry
CREATE TABLE IF NOT EXISTS users (
    device_uuid     TEXT PRIMARY KEY,
    display_name    TEXT NOT NULL,
    device_model    TEXT,
    first_seen_at   INTEGER NOT NULL,   -- Epoch seconds
    last_seen_at    INTEGER NOT NULL,   -- Epoch seconds
    current_node_id TEXT NOT NULL,      -- Serving node
    presence_status TEXT NOT NULL DEFAULT 'unconfirmed'
                    CHECK (presence_status IN ('connected', 'offline', 'unconfirmed'))
);
CREATE INDEX IF NOT EXISTS idx_users_presence ON users(current_node_id, presence_status);

-- 2. NODES: Mesh Repeater and Router Roster
CREATE TABLE IF NOT EXISTS nodes (
    node_id         TEXT PRIMARY KEY,   -- Hex identifier (e.g. '0x0001')
    name            TEXT NOT NULL,      -- Tactical callsign
    last_seen_at    INTEGER NOT NULL,
    hop_count       INTEGER NOT NULL DEFAULT 0,
    status          TEXT NOT NULL DEFAULT 'active'
);

-- 3. PERSONAL_MESSAGES: Store-and-Forward Unicast Queue (7-Day Retention)
CREATE TABLE IF NOT EXISTS personal_messages (
    message_id          TEXT PRIMARY KEY,
    sender_uuid         TEXT NOT NULL,
    recipient_uuid      TEXT NOT NULL,
    body                TEXT NOT NULL,
    original_timestamp  INTEGER NOT NULL,
    queued_timestamp    INTEGER NOT NULL,
    delivered_timestamp INTEGER,          -- NULL while pending
    status              TEXT NOT NULL DEFAULT 'pending'
                        CHECK (status IN ('pending', 'delivered', 'expired', 'failed')),
    retry_count         INTEGER NOT NULL DEFAULT 0,
    FOREIGN KEY (sender_uuid) REFERENCES users(device_uuid) ON UPDATE CASCADE,
    FOREIGN KEY (recipient_uuid) REFERENCES users(device_uuid) ON UPDATE CASCADE
);
CREATE INDEX IF NOT EXISTS idx_pm_queue ON personal_messages(status, recipient_uuid);
CREATE INDEX IF NOT EXISTS idx_pm_retention ON personal_messages(original_timestamp);

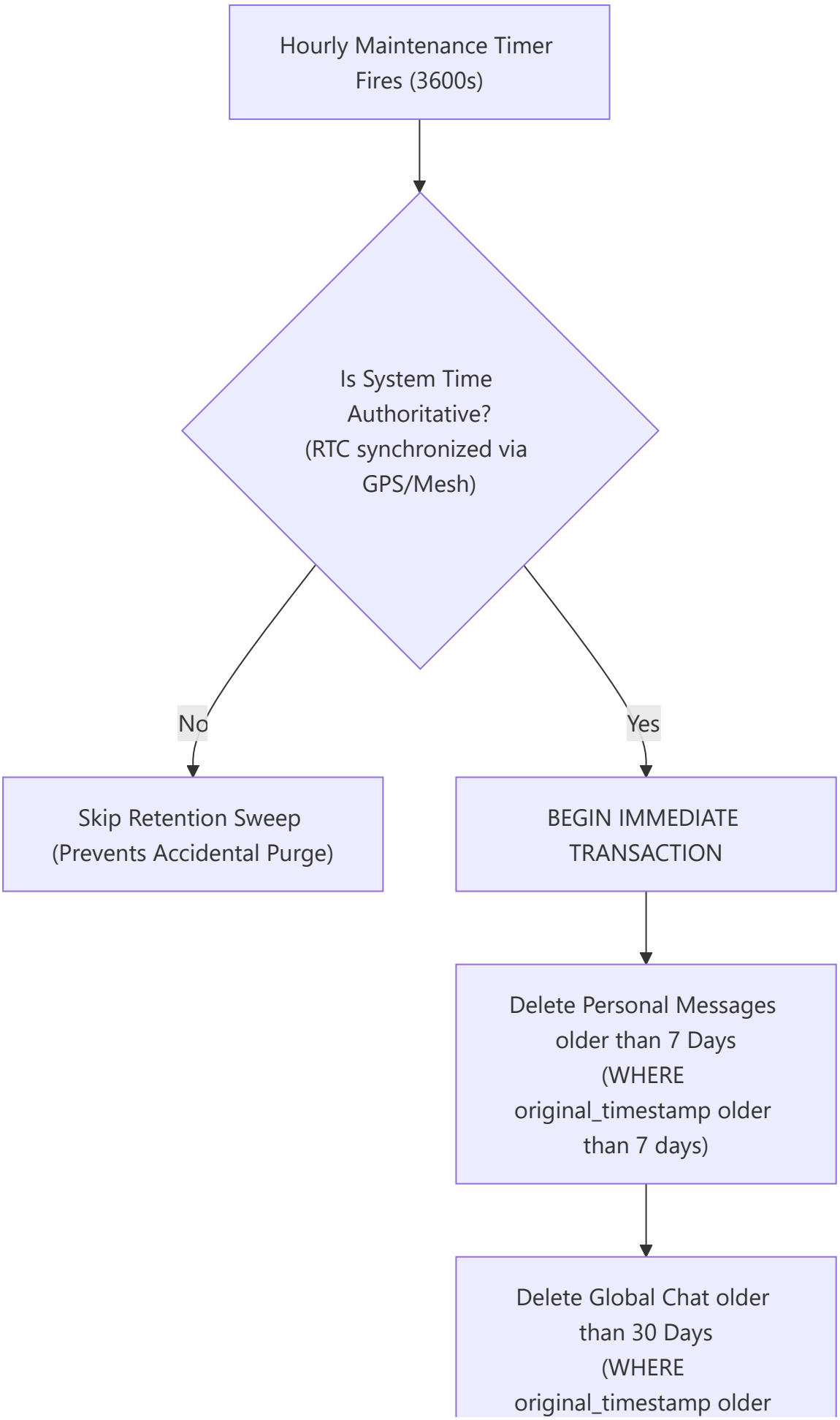
-- 4. GLOBAL_MESSAGES: Broadcast & Emergency Alerts (30-Day Retention)
CREATE TABLE IF NOT EXISTS global_messages (
    message_id          TEXT PRIMARY KEY,
    sender_uuid         TEXT NOT NULL,
    body                TEXT NOT NULL,
    original_timestamp  INTEGER NOT NULL,
    received_at         INTEGER NOT NULL,
    hop_count           INTEGER NOT NULL DEFAULT 0,
    FOREIGN KEY (sender_uuid) REFERENCES users(device_uuid) ON UPDATE CASCADE
);
CREATE INDEX IF NOT EXISTS idx_gm_retention ON global_messages(original_timestamp);

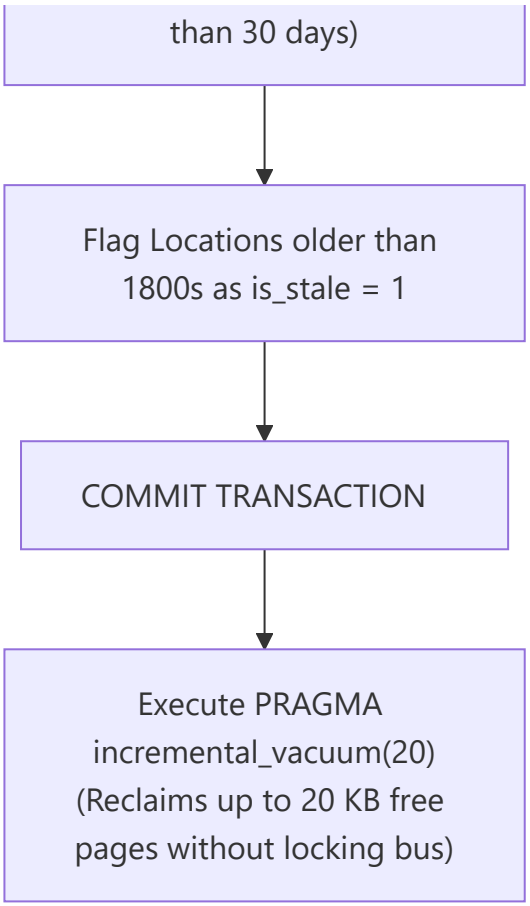
-- 5. DEVICE_LOCATIONS: Strictly 1 Row Per Device (Overwritten on update)
CREATE TABLE IF NOT EXISTS device_locations (
    device_uuid     TEXT PRIMARY KEY,
    latitude        REAL NOT NULL CHECK (latitude BETWEEN -90.0 AND 90.0),
    longitude       REAL NOT NULL CHECK (longitude BETWEEN -180.0 AND 180.0),
    altitude        REAL,
    accuracy        REAL,
    timestamp       INTEGER NOT NULL,
    is_stale        INTEGER NOT NULL DEFAULT 0 CHECK (is_stale IN (0, 1)),
    FOREIGN KEY (device_uuid) REFERENCES users(device_uuid) ON UPDATE CASCADE ON DELETE CASCADE
);

-- 6. CONFIG_AUDIT: Parameter Change Audit Trail (90-Day Retention)
CREATE TABLE IF NOT EXISTS config_audit (
    audit_id        INTEGER PRIMARY KEY AUTOINCREMENT,
    timestamp       INTEGER NOT NULL,
```

```
    changed_by      TEXT NOT NULL,
    param_name      TEXT NOT NULL,
    old_value       TEXT,
    new_value       TEXT NOT NULL
);
```

8.3 Data Retention Policies & Hourly Cleanup Lifecycle


Hourly Maintenance Timer Fires (3600s)
Is System Time Authoritative? (RTC synchronized via GPS/Mesh)
No
Yes
Skip Retention Sweep (Prevents Accidental Purge)
BEGIN IMMEDIATE TRANSACTION
Delete Personal Messages older than 7 Days (WHERE original_timestamp older than 7 days)
Delete Global Chat older than 30 Days (WHERE original_timestamp older



Record Type	Retention Policy	Purge Mechanism	Idempotent Safety Rule
User Registry	**Indefinite**	Manual administrative removal only.	Retains mesh identity mappings across extended blackouts.
Personal Messages	**7 Days** ($604,800$ s)	Hourly background task deletes rows older than 7 days from `original_timestamp`.	Incoming historical packets older than 7 days are rejected before insertion.
Global Messages	**30 Days** ($2,592,000$ s)	Hourly background task deletes rows older than 30 days from `original_timestamp`.	G-Set CRDT ensures duplicate retransmissions are ignored.
Device Location	**Latest Fix Only** (1 Row)	`INSERT ... ON CONFLICT(device_uuid) DO UPDATE` replaces previous coordinates. Marked stale if > 30 mins.	Never accumulates breadcrumb trails; protects civilian privacy.

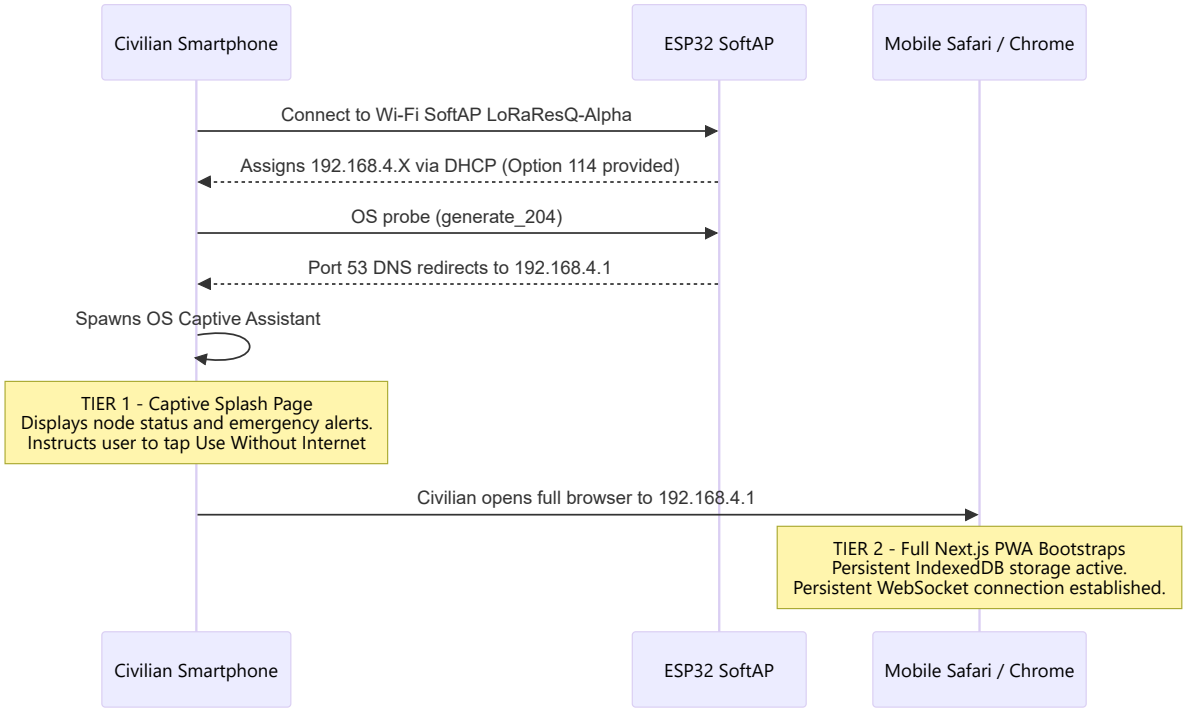
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9. Wi-Fi SoftAP, Captive Portal & Next.js PWA Client

9.1 Two-Tier Onboarding Architecture

Investigation into iOS Captive Network Assistant (CNA / `WebSheet`) and Android `CaptivePortalLogin` reveals that **sandboxed captive WebViews terminate upon dismissal, clear cookies and storage, and disable Service Workers and Geolocation**. Attempting to run the full application inside CNA leads to data loss.

LoRaResQ implements a **Two-Tier Portal Separation Pattern**:



- **Tier 1 (Captive Splash):** Lightweight inline HTML (<15 KB) served by ESP32 RAM. Shows connected node name, battery, recent broadcast alerts, a high-contrast QR code, and explicit instructions to open Safari/Chrome.
- **Tier 2 (Full Browser Console):** The user navigates to `http://192.168.4.1` in Safari or Chrome. The Next.js static SPA loads from cache, connects to `/ws`, and maintains full message history in IndexedDB.

9.2 Non-Secure Context Reality (`window.isSecureContext === false`)

Because the ESP32 operates off-grid without internet connectivity, valid public CA SSL certificates (Let's Encrypt) are impossible, and self-signed certificates trigger browser warning screens that panic disaster victims while still being denied secure-context privileges.

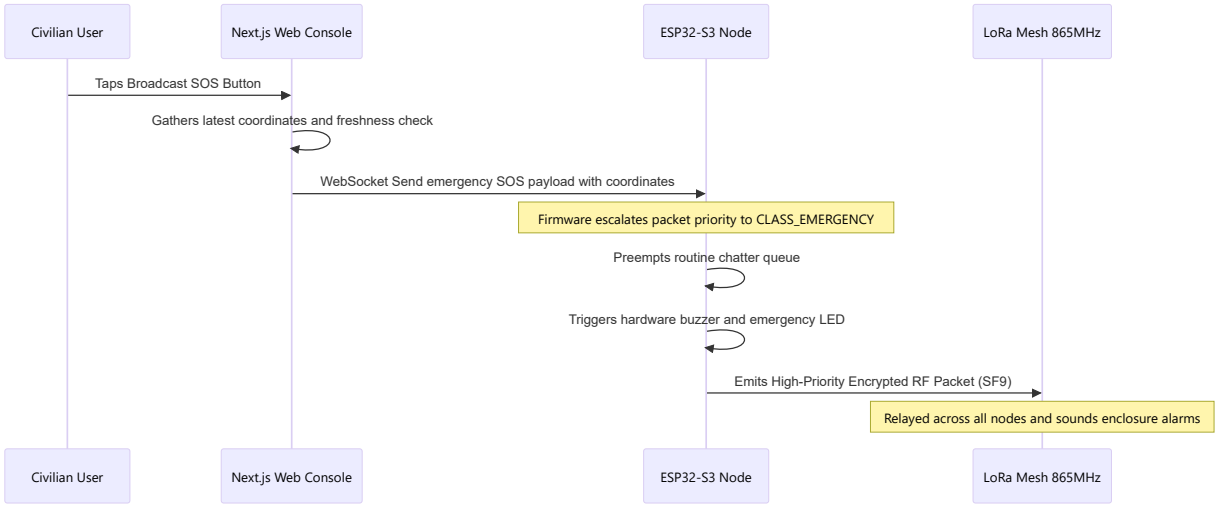
The architecture operates over plain HTTP over WPA2-PSK:

- **Geolocation:** `navigator.geolocation` is blocked on HTTP. LoRaResQ implements three fallbacks: (1) One-tap attachment of the ESP32 node's hardware GPS fix, (2) Offline interactive map pin-drop, and (3) Structured rural landmark dropdown.
- **Service Workers:** Service workers cannot register over HTTP LAN IPs. Offline persistence is achieved through HTTP `Cache-Control: public, max-age=31536000, immutable` on content-hashed static Next.js assets, paired with client-side IndexedDB persistence.
- **Device Identity:** `crypto.randomUUID()` fails on HTTP. LoRaResQ implements an HTTP-safe RFC 4122 v4 generator using `crypto.getRandomValues()` (which remains permitted in non-secure contexts), persisted across `localStorage` and `IndexedDB`.

---

10. Emergency SOS, GPS & Alert Dispatch

10.1 Web Console SOS & Location Flow



1. **Location Freshness:** Every coordinates fix carries an `updatedAt` timestamp. If a fix is older than 15 minutes, the console badges it as **"Stale Fix"**, warning rescue coordinators that the victim may have moved.
2. **5-Minute Periodic Updates:** When the browser tab is open and location access is granted, the PWA updates coordinates every 5 minutes and sends the latest fix to the node. The node **strictly overwrites** the single row in `device_locations`.

10.2 Hardware Physical SOS Button

Every physical LoRaResQ enclosure features a prominent tactile push-button wired to **GPIO 0 / 14** (debounced via FreeRTOS software timer, 50 ms):

- **Operation Without a Phone:** If a civilian has no phone or a dead battery, pressing the enclosure button for 2 seconds generates an emergency broadcast.
- **Predefined Message:** Emits firmware-configured emergency text: `"HARDWARE SOS TRIGGERED AT NODE [NodeID]"`.
- **Location Payload:** Attaches the node's internal GNSS coordinates, or the most recent valid civilian coordinates stored on the node, or explicitly flags coordinates as `"UNAVAILABLE"`. Invented or zero coordinates are strictly forbidden.

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11. Node Runtime Administration & Security

11.1 Administration Authentication & Live Parameters

The PWA provides an administrative console modal accessible via the gear icon:

- **Authentication:** Requires an administrative password provisioned at firmware flash time (default: `admin123`). The password is hash-validated on the ESP32; session tokens expire after 15 minutes of inactivity.
- **Runtime-Editable Parameters (No Reflash Required):**
  - Node Tactical Name / Callsign.
  - Radio Channel selection (Channel 15, 16, or 17).
  - RF Transmit Power (14 dBm, 17 dBm, 20 dBm, 22 dBm).
  - Wi-Fi SoftAP SSID and Passphrase.
  - Presence Heartbeat interval.
- **Audit Logging:** Every administrative modification is committed to the SQLite `config_audit` table with timestamp and parameter delta.
- **Hardware Parameters Requiring Reflash:** Primary SPI bus pinouts, partition scheme, and master AES-128 root encryption keys.

11.2 Cryptographic Security Architecture

- **Wi-Fi Link Security:** Protected by standard **WPA2-PSK** encryption.
- **LoRa Mesh Link Security:** Protected by **mbedTLS hardware-accelerated AES-128-GCM** executed inside the ESP32-S3. Over-the-air packets contain only authenticated ciphertext and a 16-byte Poly1305/GCM tag.

- **Anti-Replay Protection:** Every transmission includes a 12-byte unique nonce constructed from the monotonic message sequence counter and source node ID. Overheard replayed packets are identified and dropped before application processing.

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12. Bill of Materials (BOM) & Specifications

Component	Technical Specification	Functional Role	Est. Cost (INR)
ESP32-S3 Dev Board	Dual-core 32-bit Xtensa LX7 @ 240 MHz, 512KB SRAM, USB-C	Master controller, SoftAP host, crypto engine	₹550
EBYTE E22-900T22D-V2	SX1262 core, DIP-8, 2.54 mm pitch, SMA female, 22 dBm max	Long-range 865–868 MHz LoRa transceiver	₹850
TX868-JKD-20 Antenna	868 MHz class, 3 dBi gain, SMA-J male whip, VSWR ≤ 1.5	Matched omnidirectional RF radiator	₹180
523450 LiPo Cell	3.7V nominal, 1500 mAh capacity, lithium-polymer	Portable energy storage	₹350
MicroSD SPI Module	SPI interface with onboard 3.3V LDO regulator, 5V input	Local SQLite 3 database storage	₹90
MicroSD Card (16 GB)	Class 10 / Industrial grade FAT32 storage	Message queues, user registry, static web assets	₹320
TP4056 Type-C Charger	1A linear Li-ion charger with DW01 + 8205A dual protection	Safe USB-C battery charging & low-voltage cutoff	₹45
MT3608 Boost Module	2A DC-DC step-up converter with multi-turn trimmer	Steps 3.0–4.2V battery to steady 5.00V system bus	₹60
Decoupling Capacitors	$470\ \mu\text{F}$ (16V) low-ESR electrolytic + $100\ \text{nF}$ ceramic	Buffers transient current spikes during RF/Wi-Fi	₹15
Tactile SOS Switch	12 mm momentary push button with red cap	Enclosure hardware emergency SOS trigger	₹15
Power Switch	Miniature SPDT slide switch (rated $\geq 1\text{A}$)	Hard power disconnect	₹10
Active Buzzer & LED	5V active buzzer + 2N2222 transistor + 5mm status LED	Audible and visual SOS alarm indicators	₹35
TOTAL PER NODE	—	—	**₹2,520**
TWO-NODE PILOT TOTAL	Two fully interoperable nodes	Meets ₹10,000 student budget ceiling	**₹5,040**

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13. Assembly & Electrical Safety Protocol

13.1 Critical Assembly Directives

1. **Calibrate MT3608 Voltage BEFORE Connecting Electronics:**
   - Connect a test power supply or battery across MT3608 `VIN+` and `VIN-`.
   - Measure `VOUT+` and `VOUT-` with a digital multimeter on DC voltage mode.
   - Adjust the multi-turn brass trimmer counter-clockwise until output measures **strictly 5.00V DC**.

- *Never connect the ESP32, E22 radio, or MicroSD module to the boost output until 5.00V is verified.*

2. **Mandatory Antenna Pre-Connection Rule:**
   - **Never apply power to the node without the TX868-JKD-20 antenna securely tightened to the SMA socket.**
   - Transmitting into an open RF port generates high reflected power (VSWR $> 10:1$) that destroys the SX1262 power amplifier within milliseconds.
3. **Verify TP4056 Dual Protection ICs:**
   - Inspect the TP4056 module under magnification to confirm the presence of both the **DW01** (6-pin IC) and **8205A** (8-pin dual MOSFET). Modules lacking these chips do not provide low-voltage cutoff.
4. **Battery Soldering Sequence:**
   - Solder battery leads **last**, after all board wiring, headers, and switches are assembled. Keep bare leads insulated with Kapton tape during handling.

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14. Design Decisions & Open Questions

14.1 Key Architecture Decisions Made

1. **PWA over Flutter:** Superseded the native Flutter mobile application with a Next.js Progressive Web App (PWA). Eliminates app store download dependencies in disaster zones while supporting zero-install access across all mobile operating systems.
2. **Two-Tier Portal Separation:** Solved iOS/Android captive portal webview sandboxing by using the captive assistant strictly as a lightweight splash launchpad that directs users into their full mobile browser.
3. **Passive Overhearing over Sliding-Window:** Rejected Go-Back-N sliding windows on LoRa due to half-duplex blindness and broadcast storms. Adopted Stop-and-Wait with Passive Overhearing for intermediate hops.
4. **SQLite on MicroSD:** Adopted an embedded SQLite 3 engine on MicroSD (1 KB page size, 32-page cache, `TRUNCATE` journal) to support indexed user rosters and store-and-forward queues.
5. **Strict 1-Row Location Overwrite:** Mandated that location coordinates overwrite the single latest entry per device, preventing civilian tracking trails.

14.2 Remaining Decisions Required Before Field Deployment

Decision	Investigation Required	Responsible Role
E22 UART vs Bare SPI SX1262	Test whether EBYTE E22 UART latency satisfies sub-second emergency ACK requirements; verify if transitioning to direct SPI SX1262 is required before field pilots.	Embedded Firmware Lead
Field Pilot Terrain Testing	Execute real-world range testing in rural Maharashtra terrain across dense foliage and elevation gaps to establish empirical link budgets.	Team Lead & Mentors
Localized Dialect Presets	Finalize multi-lingual emergency preset text (Marathi, Hindi, English) in coordination with local disaster response volunteers.	Documentation & UI Lead

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15. Glossary & Academic References (Google Scholar)

15.1 Technical Glossary

Term / Acronym	Definition & Operational Context in LoRaResQ
ACK (Acknowledgement)	A sequence-matched confirmation frame emitted by the addressed destination node or intermediate relay confirming receipt of a specific packet ID.
Airtime (Time-on-Air / ToA)	The duration (in milliseconds) an RF packet physically occupies the wireless channel, governed by Spreading Factor, Bandwidth, Coding Rate, and payload bytes.

ALOHA Channel Limit	Theoretical maximum channel throughput under uncoordinated random transmission (18.4% for pure ALOHA; 36.8% for slotted ALOHA). LoRa networks operate under this limit to avoid congestion collapse.
APC (Adaptive Power Control)	Algorithmic adjustment of RF output power based on link budget and SNR to minimize energy use and reduce co-channel interference.
CAD (Channel Activity Detection)	An SX1262 hardware mode that senses the radio channel for LoRa preamble chirps in $\sim 2 \times T_{sym}$, providing carrier sensing without full demodulation.
Captive Portal	A network onboarding gateway that intercepts initial HTTP/DNS requests on unauthenticated Wi-Fi networks and redirects the browser to an onboarding page (`192.168.4.1`).
CNA (Captive Network Assistant)	The sandboxed webview environment launched automatically by iOS (`WebSheet.app`) or Android (`CaptivePortalLogin`) upon associating with a captive Wi-Fi network.
CR (Coding Rate)	The proportion of non-redundant bits in forward error correction ($4/5$, $4/6$, $4/7$, $4/8$). CR 4/5 offers a 25% overhead protection against impulse fading.
CSMA/CA	Carrier Sense Multiple Access with Collision Avoidance: a protocol in which a node listens before transmitting (using CAD or RSSI) to reduce collision probabilities.
CSS (Chirp Spread Spectrum)	LoRa's physical modulation technique, which encodes symbols using linear frequency-modulated chirps across the channel bandwidth, providing high immunity to multipath fading and Doppler shifts.
DTN (Delay-Tolerant Networking)	A network architecture designed to handle long delays, frequent disconnections, and lack of end-to-end paths through persistent store-and-forward custody transfer.
Duty Cycle	The fraction of time a radio device is permitted to transmit on a given frequency within a rolling time window (e.g., $\leq 1.0\%$ or $\leq 2.5\%$).
EIRP	*Effective Isotropic Radiated Power*: total radiated power referenced to a theoretical isotropic antenna (0 dBi). Formula: $\text{EIRP (dBm)} = P_{\text{conducted}} - L_{\text{cable}} + G_{\text{antenna}}$.
ERP	*Effective Radiated Power*: radiated power referenced to an ideal half-wave dipole (0 dBd $= 2.15$ dBi). Regulatory limit in India: 1.0W ERP (30 dBm).
G-Set CRDT	*Grow-Only Set Conflict-Free Replicated Data Type*: an append-only monotonic data model used for global chat messages, guaranteeing eventual consistency across network partitions without central arbitration.
Horizon Vector	A lightweight version vector recording the highest contiguous sequence number a node has witnessed from every peer in the mesh, used for low-bandwidth partition healing.
LBT (Listen-Before-Talk)	A medium access etiquette where the transceiver checks channel clear threshold (via CAD or RSSI) prior to keying the power amplifier.
LittleFS	A resilient, power-fail-safe, wear-leveling embedded flash filesystem developed for microcontrollers, used in LoRaResQ for static web assets.
mbedTLS	An open-source, embedded cryptographic library used in ESP-IDF to execute hardware-accelerated AES-128-GCM encryption and Poly1305 authentication.
NFAP	*National Frequency Allocation Plan*: India's statutory spectrum roadmap published by the Wireless Planning & Coordination (WPC) Wing of the Ministry of Communications.
Passive ACK (Overhearing)	An implicit hop-level delivery confirmation achieved when a transmitting node overhears its downstream neighbor relaying the exact message ID with a decremented hop count.

PWA (Progressive Web App)	A web application built with modern HTML5, CSS, and JavaScript that offers standalone installation, offline caching, and responsive UI without requiring native app store distribution.
RSSI	*Received Signal Strength Indicator*: measurement of the total received radio frequency power present in a received packet, expressed in negative dBm.
SF (Spreading Factor)	The number of chirps used to encode each symbol in LoRa (2^{SF} chips per symbol, range SF7 to SF12). Higher SF increases receiver sensitivity and range at the cost of airtime.
SNR	*Signal-to-Noise Ratio*: the ratio of received signal power to ambient RF noise power, expressed in decibels (dB). In LoRa, signals can be demodulated up to $-20\ \mathrm{dB}$ below the noise floor.
SoftAP	*Software-enabled Access Point*: an operating mode where the ESP32 acts as an autonomous 802.11 b/g/n wireless access point for up to 16 concurrent client devices.
Store-and-Forward	An asynchronous routing paradigm where packets addressed to unreachable destinations are persisted on non-volatile media (MicroSD) until the recipient reconnects.
TTL (Time-to-Live / Hop Limit)	A packet header field decremented by 1 at each intermediate relay to prevent routing loops and bound network congestion.
WPC	*Wireless Planning and Coordination Wing*: the national radio regulatory body under the Department of Telecommunications (DoT), Ministry of Communications, Government of India.

15.2 Academic References (Direct Google Scholar Citations)

The architectural, protocol, and persistence designs in this blueprint are grounded in peer-reviewed scientific literature and official telecommunications standards. Direct citation links to Google Scholar are provided below for academic verification and literature review.

A. LoRa Physical Layer, Scalability & Modulation Physics

1. **Augustin, A., Yi, J., Clausen, T., & Townsley, W. M. (2016).**
   A Study of LoRa: Long Range & Low Power Networks for the Internet of Things.
   Sensors (MDPI), 16(9), 1466. DOI: 10.3390/s16091466.
   Evaluates Chirp Spread Spectrum (CSS) modulation, receiver sensitivity, time-on-air, and link budgets.
   👉 **Google Scholar Citation**

2. **Bor, M., Roedig, U., Voigt, T., & Alonso, J. M. (2016).**
   Do LoRa Low-Power Wide-Area Networks Scale?
   In *Proceedings of the 19th ACM International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM '16)*, pp. 59–67. DOI: 10.1145/2988287.2989163.
   Analyzes channel capacity limits, pure ALOHA collision behavior, and interference under dense deployments.
   👉 **Google Scholar Citation**

3. **Centenaro, M., Vangelista, L., Zanella, A., & Zorzi, M. (2016).**
   Long-Range Communications in Unlicensed Bands: The Promise of LoRa.
   IEEE Wireless Communications, 23(4), pp. 15–21. DOI: 10.1109/MWC.2016.7721739.
   Comprehensive survey of Sub-GHz ISM operations, co-channel interference, and link budgets.
   👉 **Google Scholar Citation**

B. Multi-Hop Wireless Mesh & Controlled Flooding Protocols

4. **Cotrim, J. R., & Kleinschmidt, J. H. (2020).**
   LoRaWAN Mesh Networks: A Review and Classification of Multihop Communication.
   Sensors (MDPI), 20(15), 4273. DOI: 10.3390/s20154273.
   Evaluates multi-hop topologies, controlled flooding vs. tree routing, and half-duplex relay bottlenecks.
   👉 **Google Scholar Citation**

5. **Lee, H. C., & Ke, K. H. (2018).**
Monitoring of Large-Area IoT Sensors Using a LoRa Wireless Mesh Network System.
IEEE Sensors Journal, 18(17), pp. 7177–7183. DOI: 10.1109/JSEN.2018.2857039.
Demonstrates multi-hop LoRa packet forwarding, routing tables, and delivery ratios across complex terrain.
👉 **Google Scholar Citation**

6. **Abrardo, A., & Pozzebon, A. (2019).**
A Multi-Hop LoRa Linear Sensor Network for the Monitoring of Underground Environments.
Sensors (MDPI), 19(20), 4520. DOI: 10.3390/s19204520.
Studies multi-hop packet loss, retransmission latency, and airtime optimization in obstructed settings.
👉 **Google Scholar Citation**

C. Disaster Resilience & Emergency Ad-Hoc Communications

7. **Sciullo, L., Trotta, A., & Di Felice, M. (2020).**
Design and Performance Evaluation of a LoRa-Based Mobile Mesh Network for Disaster Scenarios.
In *IEEE Global Communications Conference (GLOBECOM '20)*, pp. 1–6. DOI: 10.1109/GLOBECOM42002.2020.9322238.
Direct architectural reference: evaluates mobile LoRa mesh nodes deployed for search-and-rescue communication.
👉 **Google Scholar Citation**

8. **Ray, P. P., Mukherjee, M., & Shu, L. (2017).**
Internet of Things for Disaster Management: State-of-the-Art and Future Challenges.
IEEE Access, 5, pp. 18818–18835. DOI: 10.1109/ACCESS.2017.2752174.
Synthesizes architectural requirements for decentralized disaster notification, telemetry, and resilience.
👉 **Google Scholar Citation**

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